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SHORT ANSWER TEST

1) Computer vision is a branch of AI that enables computers to capture, process, analyze and understand images and videos. It extracts meaningful information from visual data and makes decisions similar to human vision.

a) Main objectives of Computer vision:

- * Image understanding
- * Image classification
- * scene analysis

b) Two - real-world Applications:

- * Facial recognition systems
- * self-driving cars.

2) Computer vision provides automation image analysis using feature extraction, pattern recognition, object detection, segmentation and classification.

a) Feature extraction:

Feature extraction is the process of identifying important characteristics such as edges, corners, textures, and shapes from an image.

b) How does object recognition improve CV?

It identifies and classifies objects accurately making systems useful for surveillance, robotics, healthcare and autonomous vehicles.

- 3) Image Processing is divided into low-level, mid-level and high-level operations.
- a) Low-level processing: It involves image acquisition, enhancement, filtering and noise removal.
 - b) Difference
 - Mid-level: Image segmentation, feature extraction, object detection.
 - High-level: Image interpretation, scene understanding and decision-making.

- 4) A digital image consists of a grid of pixels, where each pixel stores brightness or colour information.
- a) Pixel: A pixel is the smallest element of a digital image representing one point of colour or intensity.
 - b) Effect of image resolution: High resolution provides more detail and better image detail, while colour resolution reduces clarity.

- 5) Image sensing: Image sensing captures light from a real-world scene and converts it into digital signals.
- a) Role of an image sensor: It converts light into electrical signals that are processed into digital images.

b) Image acquisition devices:

- 1) Digital Camera
- 2) Webcam

Sampling ∴ Sampling converts a continuous image into discrete pixels, while quantization assigns intensity values to those pixels.

Sampling ∴ Sampling is the process of dividing an image into a grid of pixels.

Purpose of Quantization ∴ It converts continuous intensity values into discrete digital values for computer processing.

Image Formation Model Working ∴

The image formation model explains how light from a scene is projected through a camera lens to create an image.

a) Perspective projection ∴ It projects a 3D object onto a 2D image while preserving depth relationships.

b) Role of Illumination ∴ Illumination provides light necessary for capturing clear images and affects images brightness and quality.

8) Image processing improves image quality, while computer vision interprets image content.

a) Image processing ∴ Image processing is the technique of enhancing, restoring and transforming digital images.

b) Two Differences ∴

- 1) Image Processing focuses on image enhancement, CV focuses on understanding images.
- 2) Image Processing produces improved images, CV produces information and decisions.

9) Applications of Computer Vision:

- a) Healthcare: CV helps detect diseases from X-rays, CT scans, MRI images and assists doctors in diagnosis.
- b) Autonomous Vehicles: It detects roads, traffic signs, pedestrians and obstacles to enable safe self-driving.

10) Digital Image fundamental provided the basic structure required for image analysis.

- a) Because every digital image is made up of pixels, each storing color and intensity information.

- b) Higher sampling and resolution produces clearer images with more details, leading to more accurate feature extraction and object recognition.